Antibiotic Prophylaxis for the Prevention of Bacterial Infections in Cirrhotic Patients With Gastrointestinal Bleeding: A Meta-Analysis

Brigitte Bernard,[1] Jean-Didier Grangé,[2] Eric Nguyen Khac,[1] Xavier Amiot,[2] Pierre Opolon,[1] and Thierry Poynard[1]

In cirrhotic patients with gastrointestinal bleeding, antibiotic prophylaxis decreases the incidence of infections but most randomized trials have not shown an increase in survival. The aim of this meta-analysis was to assess the efficacy of antibiotic prophylaxis in the prevention of infections and its effect on survival rate in cirrhotic patients with gastrointestinal bleeding. Four end points were assessed: infection, bacteremia and/or spontaneous bacterial peritonitis (SBP), incidence of SBP, and death. For each end point, heterogeneity and treatment efficacy were assessed by Der Simonian and Peto methods. Five trials including 534 patients, 264 treated with antibiotic prophylaxis for 4 to 10 days and 270 without, were identified. Mean follow-up was 12 days. Antibiotic prophylaxis significantly increased the mean percentage of patients free of infection (32% mean improvement rate, 95% confidence interval [CI]: 22-42, $P < .001$), bacteremia and/or SBP (19% mean improvement rate, 95% CI: 11-26, $P < .001$), and SBP (7% mean improvement rate, 95% CI: 2.1-12.6, $P = .006$). Antibiotic prophylaxis also significantly increased the mean survival rate (9.1% mean improvement rate, 95 % CI: 2.9-15.3, $P = .004$), without significant heterogeneity. In cirrhotic patients with gastrointestinal bleeding, short-term antibiotic prophylaxis significantly increases the mean percentage of patients free of infection and significantly increases short-term survival rate. (Hepatology 1999;29:1655-1661.)

Bacterial infections are frequently diagnosed in patients with cirrhosis and gastrointestinal bleeding. Prospective studies have shown that bacterial infections are documented in 22% of such patients within the first 48 hours after admission.[1-3] Within 7 to 14 days after initial bleeding, the incidence of bacterial infections reaches 35% to 66% among studies.[3-9] Moreover, infections seem to be closely related to prognosis in bleeding cirrhotic patients. One study has shown that bacterial infections are the major factor predictive of rebleeding within 7 days.[3] A recent study showed that bacterial infection and the use of empirical antibiotic treatment are independently associated with failure to control variceal bleeding during the first 5 days.[4]

Enteric bacteria are the most common agents of infection in cirrhotic patients.[1,10,11] A first randomized clinical trial (RCT) showed in 1985 that nonabsorbable antibiotics decreased the incidence of infection in bleeding patients.[5] Four other trials using quinolones alone or associated antibiotics have been performed in the same way.[5-9] Although these five RCTs have shown that the use of antibiotic prophylaxis decreases the incidence of infections,[5-9] only one trial has shown an increase in survival rate during the first 10 days without significant difference for mortality rate during the whole hospitalization.[5] The aim of the present meta-analysis was to assess the efficacy of antibiotic prophylaxis in the prevention of infections and its effect on survival rate in cirrhotic patients with gastrointestinal bleeding.

PATIENTS AND METHODS

Methods

Meta-analyses were performed according to a predetermined protocol based on the recommendations of Sacks et al.,[12] but were not limited to English language RCTs. Four events were chosen as end points to estimate clinical efficacy: incidence of infections, incidence of bacteremia and/or spontaneous bacterial peritonitis (SBP), incidence of SBP, and death.

Literature Search

A search of MEDLINE and a manual search were combined because we have previously shown that a MEDLINE search alone is not sensitive enough.[13] General reviews, references of published RCTs, letters to pharmacological companies, and Current Contents were also used. RCTs published in English, French, German, and Spanish were included.

Inclusion Criteria

To be included in this meta-analysis, an RCT had to fulfill the following criteria: it had to be published as an abstract or article, be randomized and prospective, include patients with cirrhosis and gastrointestinal bleeding, and assess the efficacy of antibiotic prophylaxis in the prevention of bacterial infection.

Noninclusion Criteria

Controlled trials without randomization or that compared two treatments were excluded. RCTs including patients without bleeding were also excluded.

Range of Patient Characteristics, Liver Disease, and Treatment

The following items were provided for patients: mean age, prevalence of alcohol-induced cirrhosis, Child's Pugh score,[14] prevalence of variceal bleeding, mean duration of follow-up, and number

Abbreviations: RCT, randomized clinical trial; SBP, spontaneous bacterial peritonitis; CI, confidence interval.

From the [1]Service d'Hépato-Gastroentérologie, Hôpital Pitié-Salpêtrière, Paris, France; and the [2]Service d'Hépato-Gastroentérologie, Hôpital Tenon, Paris, France.

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Address reprint requests to: Brigitte Bernard, M.D., Service d'Hépato-Gastroentérologie, Hôpital Pitié-Salpêtrière, 75651 Paris cedex, France. E-mail: bbernard@teaser.fr; fax: (33-1) 45-86-20-22.

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of patients excluded after randomization. For each study, the following items were provided: inclusion criteria for patients, exclusion criteria, end points, dosage, and duration of treatment.

Selection and Data-Extraction Bias

All RCTs considered for inclusion were analyzed independently by two observers (B. Bernard and T. Poynard). Inclusion or exclusion was not dependent on results.

Source of Support

This meta-analysis was not supported by any pharmaceutical company, government agency, or other grants.

Statistical Methods

In the selected RCTs, exclusion criteria were defined in the protocol study. Therefore, some patients were randomized and excluded after randomization according to these protocols. These patients were not included in the meta-analysis.

Criteria for Combinability

The following methods were used to assess combinability: comparison of each end point improvement in control groups by the χ^2 test and heterogeneity tests by the Peto method[15] and by the Der Simonian and Laird method.[16] The Peto method is a variant of the Mantel-Haenszel test. The hypothesis states that there is an overall treatment effect (fixed model). In each study, a quantification of the size of the difference observed between the 2 groups (*i.e.*, antibiotic prophylaxis and no treatment) is performed using the odds ratio (Odds Ratio = Exp[(Observed − Estimated)/Variance(Observed − Estimated)] with an estimate of the 95% confidence interval (CI). An estimation of the overall size of the difference between the groups is then performed with its CI, with a graphic representation, a test of the statistical significance of the overall difference, and a test of the homogeneity of the results obtained in the individual studies. For the Der Simonian and Laird method, the hypothesis is that the treatment effect is not fixed (random model). The estimate is the difference of the percentages (risk of treated patients − risk of controls). The Der Simonian and Laird method is less sensitive to heterogeneity and more conservative than the Peto method. These two methods were performed for each analysis and all significant results were concordant. Because of the lack of power of heterogeneity testing for the detection of insignificance but still substantial heterogeneity, two sensitivity analyses were performed.

For each evaluated end point, the following strategy was used: (1) heterogeneity was assessed between results in the control groups, (2) the efficacy of antibiotic prophylaxis versus no treatment was assessed by the Peto method, and (3) an assessment was made by the Der Simonian and Laird method. When a significant difference was observed between the two treatments with the two latter methods, sensitivity analyses were performed twice by the Peto and Der Simonian stratified meta-analyses: (1) after exclusion of the RCT using oral, nonabsorbable antibiotics, (2) after exclusion of RCTs using intravenous antibiotics, and (3) after inclusion of patients excluded. This sensitivity analysis was performed to minimize a type 1 error and to avoid finding a difference when none existed. In this sensitivity analysis, patients excluded after randomization were considered to have bacterial infections and to have died. The methodological quality was assessed for each RCT by two observers independently using a previously validated questionnaire.[17] Detailed results of stratified meta-analyses were provided using the Der Simonian method. Results of heterogeneity tests were only provided when significant.

A significance level of 5% was taken as the alpha risk. Each estimate was given with its 95% CI. Comparison of odds ratios of rates of differences between strata were performed using their 95% CIs. When the percentage of an end point was 0% or 100% in both groups, relative risks and difference rates were impossible to compute directly by the Peto or Der Simonian methods. Therefore in these cases, a pseudocount method was used: 0.5 was added to the number of subjects in each group; this method does not change the ratio (equal to one), and only slightly reduced the CI.

For each end point and each RCT, detailed results were also provided in figures to compare the improvement rates between treatment groups and for meta-analysis by the Der Simonian and Laird method. In these figures, an insignificant homogeneity test ($P > .05$) indicated that there was no significant heterogeneity.

RESULTS

Selected RCTs

A total of six randomized trials were identified between 1985 and 1998.[5-9,18] Among these trials, one was excluded: it was published as abstract and the number of patients with infection and other events was not available.[18] Thus, a total of five RCTs comparing antibiotic prophylaxis and no treatment were selected for the meta-analysis of the efficacy of antibiotic prophylaxis in the prevention of bacterial infection and for the effect on survival in cirrhotic patients with gastrointestinal bleeding.[5-9]

Design of Selected RCTs and Patient Characteristics

The methodology of each selected RCT and the clinical characteristics of included patients are given in Tables 1 and 2. All patients had cirrhosis and gastrointestinal bleeding. In one RCT, only patients considered with a high risk of infection (Child-Pugh's class C or rebleeding) were included.[8] In another trial, 40% of patients had hepatocellular carcinoma.[9] In all RCTs, patients with patent signs of infection, and patients treated with antibiotics within previous weeks were not included.[5-9] After randomization, exclusion criteria were as follows: surgery or death within the first 24 hours,[5,6] surgery within the first 24 hours or death within the first 12 hours,[8] proven infection on admission,[7,8] or bleeding not related to esophageal varices.[7] A total of 29 patients were excluded from treated groups (14%) and 32 from control groups (15%). End points were the prevention of bacterial infections during the study period, which corresponded to the follow-up ranging from 7 to 14 days (mean ± SD, 11.5 ± 2.6 days). Only one RCT was placebo-controlled.[9] Antibiotic prophylaxis was an association of oral, nonabsorbable antibiotics (gentamycin + vancomycin + nystatin or neomycin + colistin + nystatin) in one RCT,[5] oral norfloxacin in one RCT,[6] intravenous then oral ofloxacin with intravenous bolus of amoxicillin and clavulanic acid before each endoscopy in one RCT,[7] intravenous then oral ciprofloxacin and amoxicillin and clavulanic acid in one RCT,[8] and oral ciprofloxacin in the last RCT.[9] The duration of treatment was 4 days,[5,7] 7 days,[6,9] or 10 days.[8]

Results of the selected RCTs are described in Table 3. Patients excluded after randomization were not included in the analysis. Survival was given at the end of follow-up in three RCTs,[5-7] at 28 days for one RCT,[8] and at 30 days in the last one.[9] The mean observation time for survival was 19.3 ± 8.0 days.

Adverse Events

Adverse events were mentioned in three RCTs.[6-8] In the antibiotic prophylaxis groups, four patients had diarrhea in one RCT.[7] In the control groups, no adverse events were reported.[6-8] In one RCT, no major side effects that could be attributable to antibiotics were noted.[9]

TABLE 1. Design of the Studies and Treatments

First Author	Inclusion Criteria	Exclusion Criteria	End Points	Treatment, Route, Dosage, and Duration	Mean Follow-up (days) Ab/C
Rimola 1985[5]	Cirrhosis GI bleeding	Signs of infection Antibiotic within 2 last wk Surgery or death within first 24 h	Prevention of infection within 10 days after bleeding stops	GVN (40 patients) orally NCN (34 patients) orally until 48 h after bleeding stops	10.9/10.1
Soriano 1992[6]	Cirrhosis GI bleeding	Signs of infection Antibiotic within 2 last wk Transfers from other hospitals Surgery or death within first 24 h	Prevention of infection during hospitalization Mean 14 days	Norfloxacin orally 800 mg/d for 7 days	13.5/14.4
Blaise 1994[7]	Cirrhosis GI bleeding	Signs of infection Antibiotic within 2 last wk Infection on admission No esophageal varices	Prevention of infection within first 14 days	Ofloxacin 400 mg/d IV then orally, for 10 days, and IV bolus amoxicillin + clavulanic acid 1 g before each endoscopy	14
Pauwels 1996[8]	Cirrhosis GI bleeding Child Pugh score class C or rebleeding	Signs of infection Antibiotic within last wk Infection on admission Death within first 12 h Surgery within first 24 h	Prevention of infection within 10 days after bleeding stops	Ciprofloxacin 400 mg/d + amoxicillin + clavulanic acid 3 g/d, IV then orally, until 3 days after bleeding stops	10 days after bleeding stops
Hsieh 1998[9]	Cirrhosis GI bleeding	Signs of infection Antibiotic within 2 last wk	Prevention of infection within first 7 days	Ciprofloxacin 1,000 mg/d orally for 7 days	7 days

Abbreviations: Ab, antibiotic prophylaxis group; C, control group; GI, gastrointestinal; IV, intravenous; GVN, [Gentamycin 200 mg + vancomycin 500 mg + nystatin 1 M]/6 h; NCN, [neomycin 1 g + colistin 1.5M + nystatin 1 M]/6 h.

Methodological Quality of Selected RCTs

The methodological quality of each selected RCT is shown in Table 4. Methodological score was homogeneous, ranging from 13 to 18.

Meta-analysis

Percentage of Patients Free of Infection. The five randomized trials including 534 patients, 264 treated with antibiotic prophylaxis and 270 not treated, were available for this end point.[5-9] There was significant heterogeneity in the results in the control groups ($P = .008$). The mean percentage of patients free of infection was 86% in the group of treated patients and 55% in the control group. The difference was significant in favor of antibiotic prophylaxis by the Peto method (odds ratio = 4.64, 95% CI: 3.19-6.75, $P < .001$), and by the Der Simonian method, with a mean rate difference of 32.3% (95% CI: 22.6-42.0, $P < .001$) (Fig. 1). There was no significant heterogeneity with both methods.

The results of the sensitivity analyses are described in Table 5. Meta-analysis after exclusion of the trial evaluating the use of oral, nonabsorbable antibiotics[5] showed that the difference was significant by the Peto method (odds ratio = 5.59, 95% CI: 3.63-860, $P < .001$), and by the Der Simonian method (mean rate difference: 35.8%, 95% CI: 27.6-44.0, $P < .001$). Meta-analysis after exclusion of RCTs using intravenous antibiotics[7,8] showed a significant difference by the Peto method (odds ratio = 3.99, 95% CI: 2.52-6.13, $P < .001$) and by the Der Simonian method (mean rate difference: 26.8%, 95% CI: 17.4-36.1, $P < .001$). Meta-analysis after inclusion of patients excluded in the four RCTs[5-8] showed a significant difference by the Peto method (odds ratio = 3.45, 95% CI: 2.45-4.86, $P < .001$), and by the Der Simonian method (mean rate difference: 27.8%, 95% CI: 19.2-36.5, $P < .001$).

Percentage of Patients Free of SBP and/or Bacteremia. The five RCTs were available for this end point.[5-9] There was significant heterogeneity in the results in the control groups ($P = .004$). The mean percentage of patients free of SBP and/or bacteremia was 92% in the group of treated patients and 73%

TABLE 2. Clinical Characteristics of Patients Included in Selected RCTs

First Author	Number of Patients*	Patients Excluded†	Mean Age (Years)	Alcoholic Cirrhosis (%)	Child-Pugh Score			Variceal Bleeding (%)
					A (%)	B (%)	C (%)	
Rimola 1985[5]	68/72	6/3	58/56	51/51	ND	ND	ND	88/86
Soriano 1992[6]	60/59	4/5	61/58	55/58	32/36	50/42	18/22	75/78
Blaise 1994[7]	46/45	12/14	52/54	89/87	0/0	24/20	76/80	100/100
Pauwels 1996[8]	30/34	7/10	51/53	90/62	7/0	10/29	83/71	60/71
Hsieh 1998[9]	60/60	0/0	58/62	10/15	8/10	55/52	37/38	70/68

NOTE. All values reflect antibiotic prophylaxis group/controls.

Abbreviation: ND, not determined.

*Patients considered for analysis.

†Patients randomized and excluded according to the protocol.

TABLE 3. Results of Selected RCTs

First Author	Number of Patients	Patients Excluded	Patients Free of Infections	Patients Free of Bacteremia and/or SBP	Patients Free of SBP	Survival
Rimola 1985[5]	68/72	6/3	57/47	62/57	ND	61/55
Soriano 1992[6]	60/59	4/5	54/37	58/49	58/55	56/52
Blaise 1994[7]	46/45	12/14	37/15	38/25	43/38	35/29
Pauwels 1996[8]	30/34	10/7	26/16	28/20	29/27	26/26
Hsieh 1998[9]	60/60	0/0	54/33	57/46	57/52	47/42

NOTE. All values reflect antibody prophylaxis group/controls.
Abbreviation: ND, not determined.

in the control group. The difference was significant in favor of antibiotic prophylaxis by the Peto method (odds ratio = 3.86, 95% CI: 2.45-6.06, $P < .001$) and by the Der Simonian method, with a mean rate difference of 18.5% (95% CI: 11.4-25.6, $P < .001$) (Fig. 2). There was no significant heterogeneity with both methods.

Sensitivity meta-analysis after exclusion of the trial evaluating the use of oral, nonabsorbable antibiotics[5] showed that the difference was significant by the Peto method (odds ratio = 4.39, 95% CI: 2.62-7.37, $P < .001$) and by the Der Simonian method (mean rate difference: 20.8%, 95% CI: 12.3-29.3, $P < .001$). Meta-analysis after exclusion of RCTs using intravenous antibiotics[7,8] showed a significant difference by the Peto method (odds ratio = 3.54, 95% CI: 1.95-6.40, $P < .001$), and by the Der Simonian method (mean rate difference: 14.5%, 95% CI: 8.0-21.1, $P < .001$). Meta-analysis after inclusion of patients excluded in the four RCTs[5-8] showed a significant difference by the Peto method (odds ratio = 2.48, 95% CI: 1.69-3.63, $P < .001$), and by the Der Simonian method (mean rate difference: 15.6%, 95% CI: 9.2-21.9, $P < .001$).

Percentage of Patients Free of SBP. Three RCTs including 394 patients, 196 treated with antibiotic prophylaxis and 198 not treated, were available for this end point.[6-9] There was no significant heterogeneity in the results in the control groups. The mean percentage of patients free of SBP was 95% in the group of treated patients and 87% in the control group. The difference was significant in favor of antibiotic prophylaxis by the Peto method (odds ratio = 2.84, 95% CI: 1.42-5.69, $P = .003$) and by the Der Simonian method, with a mean rate difference of 7.4% (95% CI: 2.1-12.6, $P = .006$) (Fig. 3). There was no significant heterogeneity with both methods.

Sensitivity meta-analysis after exclusion of the trial evaluating the use of oral, nonabsorbable antibiotics[5] showed the same results because data concerning SBP were not available in this trial. Meta-analysis after exclusion of RCTs using intravenous antibiotics[7,8] showed no significant difference by the Peto method (odds ratio = 2.44, 95% CI: 0.91-6.53, $P = .07$) and by the Der Simonian method (mean rate difference: 5.3%, 95% CI: 0.0-11.5, $P = .10$). Meta-analysis after inclusion of patients excluded in the three RCTs[6-8] showed significant difference by the Peto method (odds ratio = 1.64, 95% CI: 1.01-2.66, $P = .04$) and by the Der Simonian method (mean rate difference: 7.1%, 95% CI: 0.6-13.6, $P = .03$).

Survival Rate. The five RCTs were available for this end point.[5-9] There was no significant heterogeneity in the results in the control groups. The mean survival rate was 85% in the group of treated patients and 76% in the control group. The difference was significant in favor of antibiotic prophylaxis by the Peto method (odds ratio = 1.88, 95% CI: 1.22-2.89, $P = .004$) and by the Der Simonian method with a mean rate difference of 9.1% (95% CI: 2.9-15.3, $P = .004$) (Fig. 4). There was no significant heterogeneity with both methods.

Sensitivity meta-analysis after exclusion of the trial evaluating using oral, non absorbable antibiotics[5] showed that the difference was significant by the Peto method (odds ratio = 1.71, 95% CI: 1.04-2.80, $P = .03$) and by the Der Simonian method (mean rate difference: 7.6%, 95% CI: 0.4-14.8, $P = .04$). Meta-analysis after exclusion of RCTs using intravenous

TABLE 4. Questionnaire for Assessing Methodological Quality of RCTs

	Rimola 1985[5]	Soriano 1992[6]	Blaise 1994[7]	Pauwels 1996[8]	Hsieh 1998[9]
1. Is (are) the major endpoint(s) clearly stated?	2	2	2	2	2
Description of the population					
2. Are the criteria of inclusion and exclusion clearly stated?	2	2	2	2	2
3. Is the number of patients seen and rejected given?	1	0	0	2	0
4. Is the number of patients randomized clearly stated for both groups?	2	2	2	2	2
5. Is the number of patients who were withdrawn and dropped out and the reasons clearly stated?	2	2	2	2	1
Blindless of the trial					
6. Are the patients blind to treatment?	0	0	0	0	1
7. Are the physicians caring for patients blind to treatment?	0	0	0	0	0
8. Are the persons assessing major endpoints blind to treatment?	0	0	0	0	0
Statistical method					
9. Is the necessary sample size precalculated?	0	0	0	2	0
10. Is the method of randomization given and is this randomization blind?	1	0	0	1	0
11. Are pretreatment variables adequately analyzed and discussed?	2	2	2	2	2
12. Are statistical tests used appropriate?	2	2	2	2	2
13. Is handling of withdrawals correct?	1	1	1	1	2
14. If "nonsignificant" trial, the smallest sample size per group is greater or equal to 100 (0), between 30 and 100 (−1), smaller than 30 (−2)	0	0	0	0	0
Total score (−2 to 26)	15	13	13	18	14

NOTE. For the first 13 items, 2 signifies Yes, 1 Partial, and 0 No or unknown.

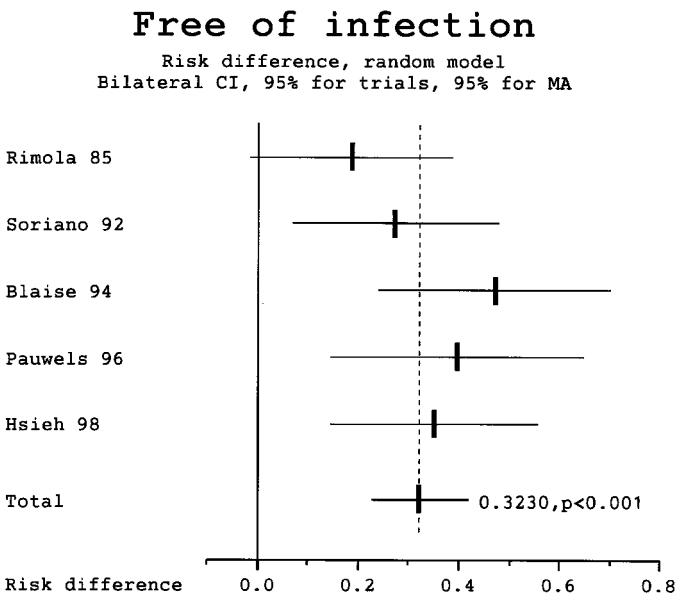


FIG. 1. Mean percentage of patients free of infection (Der Simonian).

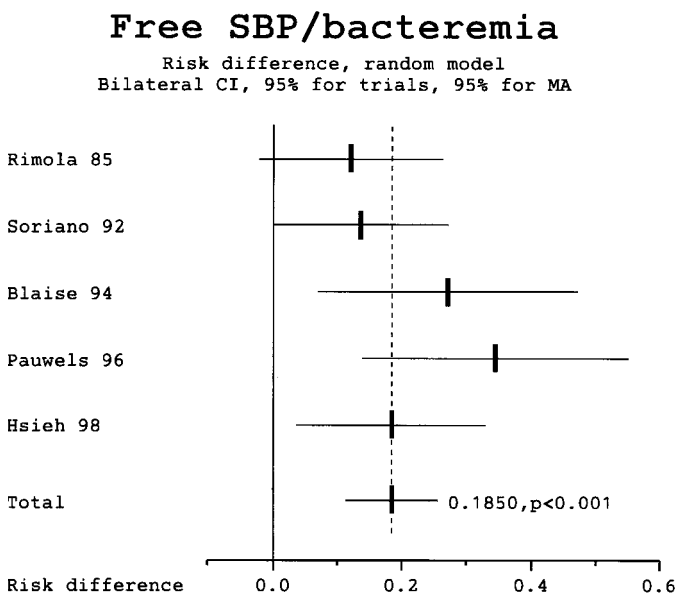


FIG. 2. Mean percentage of patients free of SBP and/or bacteremia (Der Simonian).

antibiotics[7,8] showed a significant difference by the Peto method (odds ratio = 1.92, 95% CI: 1.12-3.29, $P = .02$) and a significant difference by the Der Simonian method (mean rate difference: 8.6%, 95% CI: 1.5-15.6, $P = .02$). Meta-analysis after inclusion of patients excluded in the four RCTs[5-8] showed a significant difference by the Peto method (odds ratio = 1.50, 95% CI: 1.04-2.16, $P = .03$) and a significant difference by the Der Simonian method (mean rate difference: 7.6%, 95% CI: 0.8-14.4, $P = .03$).

TABLE 5. Results of Sensitivity Analyses (Der Simonian method)

Strata of Sensitivity Analysis Results for Each End Point	RCTs	Mean Risk Difference (95% CI)	*P*	Heterogeneity (*P*)
Exclusion of RCT using nonabsorbable antibiotics[5]				
Mean percentage of patients				
free of infection	6-9	35.8% (27.6-44.0)	<.001	NS
free of SBP and/or bacteremia	6-9	20.8% (12.3-29.3)	<.001	NS
Survival	6-9	7.6% (0.4-14.8)	.04	NS
Exclusion of RCTs using intravenous antibiotics[7,8]				
Mean percentage of patients				
free of infection	5, 6, 9	26.8% (17.4-36.1)	<.001	NS
free of SBP and/or bacteremia	5, 6, 9	14.5% (8.0-21.1)	<.001	NS
free of SBP	6, 9	5.3% (0.0-11.5)	.10	NS
Survival	5, 6, 9	8.6% (1.5-15.6)	.02	NS
Inclusion of patients excluded after randomization				
Mean percentage of patients				
free of infection	5-9	27.8% (19.2-36.5)	<.001	NS
free of SBP and/or bacteremia	5-9	15.6% (9.2-21.9)	<.001	NS
free of SBP	6-9	7.1% (0.6-13.6)	.03	NS
Survival	5-9	7.6% (0.8-14.4)	.03	NS

Abbreviation: NS, not significant.

DISCUSSION

The selected RCTs in this meta-analysis were few but homogeneous. Included patients all had cirrhosis and gastrointestinal hemorrhage. Patients treated with antibiotics and patients with patent signs of infection on admission were not included, and surgery or death within the first 24 hours were exclusion criteria in three RCTs.[5,6,8] End points were the prevention of infection within the first 7 to 14 days in all RCTs.[5-9] Antibiotic prophylaxis used fluoroquinolones in four RCTs,[6-9] associated with amoxicillin and clavulanic acid

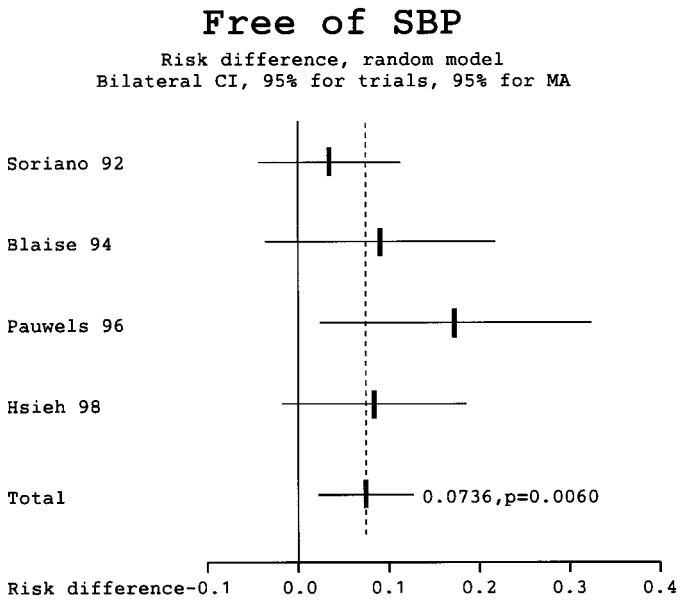


FIG. 3. Mean percentage of patients free of SBP (Der Simonian).

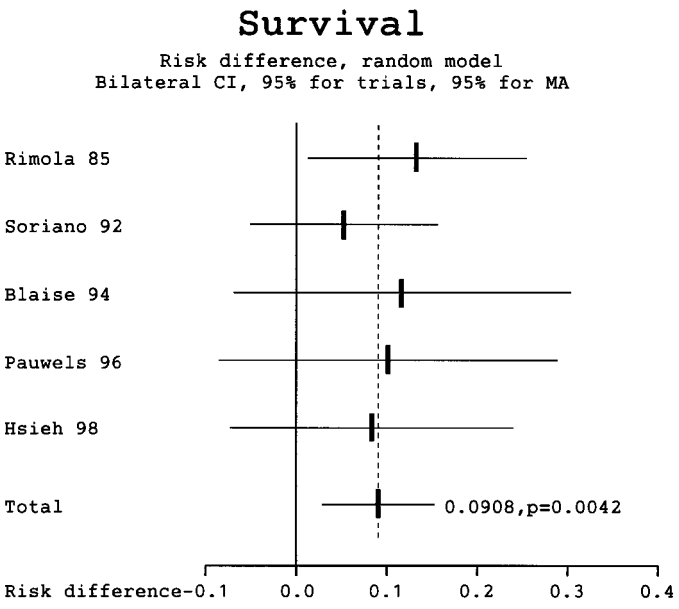


FIG. 4. Mean survival (Der Simonian).

in two of them[7,8] and oral, nonabsorbable antibiotics in one RCT.[5] Although these antibiotic regimen may appear to be different, efficacy on the prevention of infection was clearly proven in the five RCTs, including the trial that used nonabsorbable antibiotics.[5] Moreover, the two sensitivity analyses after exclusion of the trial using nonabsorbable antibiotics[5] and after exclusion of trials using intravenous antibiotics[7,8] showed similar results. This point confirms that the main source of infections in these patients remains the digestive tract.[4] The treatment duration was similar in the five RCTs, ranging from 4 to 10 days, as well as the mean follow-up from 7 to 14 days.

This meta-analysis showed a significant heterogeneity between the control groups for two end points: the percentage of patients free of infection and free of SBP and/or bacteremia. This can be easily explained by the greater severity of patients in two trials,[7,8] reflected by a high percentage of patients classified as Child-Pugh's class C (80%)[7,8] compared with 20% in the trial of Soriano et al.[6] and 38% in the trial of Hsieh et al.[9] Indeed, in the trial of Pauwels et al.,[8] only patients considered being at high risk of infection, Child-Pugh's class C, or with rebleeding were included. In the trial of Blaise et al.,[7] all patients had a central venous catheter, 35% necessitated ventilatory assistance, and more than 60% had hemorrhagic shock. Patients early operated or dead were not excluded as in three other trials.[7] The mean percentage of patients free of infection in the control group was, respectively, in the five RCTs: 65%,[5] 63%,[6] 33%,[7] 47%,[8] and 55%.[9] These discrepancies appeared similarly for the mean percentage of patients free of SBP and/or bacteremia in the control groups: 79%,[5] 83%,[6] 55%,[7] 58%,[8] and 77%.[9] Statistical heterogeneity is probably a consequence of the different liver disease severities.

Although these are significant heterogeneities between control groups, the efficacy of antibiotic prophylaxis was clearly established. Results of individual selected RCTs have shown significant differences for the incidence of infection in four RCTs[6-9] and for the incidence of SBP and/or bacteremia in the selected five RCTs.[5-9] Meta-analysis confirmed the efficacy of antibiotic prophylaxis in the prevention of infection, and was useful for estimating the level of efficacy versus no treatment. The difference of treatment efficacy seemed linked to the severity of liver disease, as shown in the figures. Indeed the mean improvement rate of patients free of infection and free of SBP and/or bacteremia was more marked in patients with more severe cirrhosis[7,8] than in patients without[5,6,9] (Figs. 1 and 2). The greater the risk of infection, the greater the benefit from a decrease infection. The efficacy of treatment is confirmed by the results of sensitivity analyses, with significant differences for all end points after exclusion of the trial using nonabsorbable antibiotics, and after inclusion of patients excluded in the trials. The sensitivity analysis after exclusion of the trial of Rimola et al.[5] showed greater differences for infection rates, suggesting that absorbable antibiotics may be more efficient. The sensitivity analysis after exclusion of trials using intravenous antibiotics[7,8] showed significant differences for all end points, except the incidence of SBP (2 RCTs), and significant differences for survival despite the small number of RCTs (3 RCTs, 379 patients).

Results of individual selected RCTs showed a significant improvement in survival in only one trial.[5] For the first time, this meta-analysis showed that short-term antibiotic prophylaxis significantly increases the mean survival rate at 19 days. There was no significant heterogeneity between treatments efficacies. The mean improvement rate in living patients was more than 9%. This difference was significant despite the small number of trials. The three sensitivity analyses confirmed the significant difference. Adverse events of treatment were missing in most patients.[5-9] The development of resistant strains in treated patients was not observed in the trials. This risk for such a short duration of treatment, between 4 to 10 days, seems to be very low.

For the first time, this meta-analysis has shown that short-term antibiotic prophylaxis significantly increases the survival rate at 19 days in cirrhotic patients with gastrointestinal hemorrhage. Indeed despite a heterogenity between control groups for two end points, the efficacy of antibiotic prophylaxis in these patients is clearly established. The mortality rate after bleeding in patients with cirrhosis is very high, close to 30% after 3 weeks. A treatment that increases survival without adverse events must be considered with attention. Moreover, one study has shown that the cost of antibiotic prophylaxis was lower than the treatment of infections .[8] For all these reasons, antibiotic prophylaxis for 7 days should be systematically used in cirrhotic patients with gastrointestinal bleeding after bacteriological samples have been performed. Lastly, for new randomized trials in these patients, a placebo group seems ethically difficult to accept.

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